Every Day Counts: Dense EUA Supply Shocks from Asset Prices

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Abstract

This paper proposes a new identification strategy for daily shocks to the supply of European Union emission allowances (EUAs). Within a daily SVAR of European energy and financial markets, we employ sign and narrative restrictions to identify one shock realization per trading day. This contrasts with the high-frequency-identified carbon policy surprise measure of Känzig (2023), which relies on only a few dozen daily realizations, limiting its usefulness in data-intensive applications such as time-varying or nonlinear analyses. At daily frequency, a restrictive EUA supply shock raises the carbon price and lowers valuations of European utility stocks on impact. Aggregated to monthly frequency and used as an external instrument, a shock that raises energy prices by 1 percent lowers greenhouse gas emissions by around 0.59 percent at peak, depresses industrial production by around 0.7 percent, and raises unemployment by over 0.1 percentage points. Whenever they overlap, our daily shock series is highly correlated with the carbon policy surprise shock. However, censoring our shock series on every regulatory announcement day underlying the carbon policy surprise measure leaves our impulse responses almost unchanged. Thus, our results indicate that EUA supply shocks are not confined to regulatory events.

Keywords: EU ETS, carbon pricing, emission allowances, sign restrictions, narrative restrictions, proxy SVAR.

JEL classification: C32, E32, G12, Q48, Q54, Q58.

*The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Deutsche Bundesbank or the Eurosystem.

1 Introduction

The instrument to identify carbon policy shocks available to macroeconomists is observed on a few dozen days (Känzig, 2023). Measuring carbon policy surprises in tight windows around regulatory announcements delivers a cleanly identified series, but a sparse one. We instead recover a shock to the supply of allowances in the EU Emissions Trading System (EU ETS) on every trading day. We estimate a daily ten-variable SVAR of European energy and financial markets and identify a restrictive EU allowance (EUA) supply shock as an innovation that raises the carbon price and lowers the valuation of European utilities. To sharpen identification, we also impose the sign of the shock on a small set of dates, so that it agrees with the established narrative account of these events. The resulting daily series is positively correlated with the high-frequency carbon policy surprise of Känzig (2023) on the days on which the latter surprise is non-zero. Used as an external instrument in a monthly proxy-SVAR, it delivers emissions and macroeconomic responses close to those from the high-frequency instrument. The structural shocks that the two proxy-SVARs recover correlate at 0.89. An EUA supply shock that raises energy prices by 1 percent lowers greenhouse gas emissions by around 0.59 percent at peak and industrial production by around 0.7 percent. Allowances are bankable, so their price reflects the expected path of scarcity, and news that shifts that path need not arrive through a scheduled regulatory announcement. Because we observe the shock on every trading day, we can also censor it on exactly the announcement days behind the high-frequency measure. The responses barely change, confirming that EUA supply shocks are not confined to the regulatory calendar.

The EU Emissions Trading System is the key policy instrument to bring down greenhouse gas (GHG) emissions in the EU. A cap limits the total amount of emissions, while firms buy emission allowances (EUAs) in auctions and trade them in secondary markets. Since 2013, the EU ETS has raised more than 175 billion euros and has become economically important for firms, asset prices, and the broader macroeconomy. In this paper we identify an EUA supply shock and quantify its impact on the European economy.

To identify the EUA supply shock, we estimate a ten-variable VAR of European energy and financial markets at daily frequency. A global stock market measure enters as an exogenous control. We identify the EUA supply shock assuming that an allowance supply shock raises carbon prices and lowers utility stock valuations. The key idea underlying our identification is that European utility firms are heavy emitters, serve predominantly local demand and were covered by the regulation from the start. Recent work also finds the power sector to play a key role in the transmission of EU ETS price changes to the economy and GHG emissions

(Känzig, 2023). On top of the sign restrictions on the impact responses, we impose the sign of the EUA supply shock itself on key historical days, so that it agrees with the established narrative account of these episodes.

Our restrictions do not isolate regulatory news alone. Any disturbance that tightens effective allowance supply relative to the abatement options available to utilities satisfies them. Therefore, the narrative restrictions tighten the identified set by discarding models in which energy price disturbances, for example, generate the required sign pattern. Four pieces of evidence point to supply-side disturbances dominating the resulting series. First, the daily point estimates line up with the event-study evidence in Bauer, Känzig, & Rudebusch (2026). Energy futures prices rise, while nominal interest rates do not respond. Second, the shock series is positively correlated with the surprise series of Känzig (2023), which is built entirely from regulatory news. Third, used as an instrument in a monthly proxy-SVAR, it reproduces the macroeconomic responses that the surprise delivers. Fourth, the responses hold up even when we drop the narrative restrictions.

Our approach has three advantages over existing studies, which estimate a carbon policy shock (e.g. Känzig (2023) or Bjørnland, Cross, & Kapfhammer (2024)). First, we recover the daily shock from a structural model. The same model can be applied to estimate the effects of carbon allowance supply shocks in other carbon markets. Second, we estimate a dense shock series. The model delivers one shock realization for each trading day in the sample, which may improve estimation and inference in downstream applications. Third, the series can be updated with little effort as new observations become available.

The remainder of the paper proceeds as follows. Section 2 places the paper in the literature on the aggregate effects of carbon pricing. Section 3 describes the EU ETS and explains why listed European utilities are the sector most exposed to allowance supply. Section 4 sets out the daily SVAR, the sign and narrative restrictions, and the resulting shock series. Section 5 aggregates the shock to monthly frequency and estimates its effects on emissions and the European macroeconomy. Section 6 asks how much of the identification the sign restrictions deliver on their own. Section 7 concludes.

2 Related Literature

A substantial body of research studies the effects of carbon pricing at the micro level. For example, Colmer, Martin, Muûls, & Wagner (2025) use a matched difference-in-differences design and find that the EU ETS induced regulated firms to reduce CO_2 emissions relative

to unregulated firms by 14% during Phase 1 and by 16% during Phase 2, with no detectable negative effects on the economic performance of firms. Their design credibly identifies the firm-level effects of the EU ETS. However, moving from these micro estimates to macro effects raises the missing-intercept problem emphasized by Moll & Hanney (2025).

Our paper is most closely related to a smaller literature identifying the aggregate effects of carbon pricing. Känzig (2023) identifies carbon policy surprises by measuring high-frequency carbon price changes in tight windows around regulatory announcements. The regulatory announcements concern the supply of emission allowances. Identification relies on the assumption that regulatory news dominates carbon price movements within a tight window, while other shocks are predetermined with respect to the announcements. A growing literature uses the resulting carbon policy surprise, including Ortubai, Piacentini, Serrador, & Rubio-Ramırez (2025), A. G. Gazzani & Taboga (2024), Bauer et al. (2026), Konradt & Mangiante (2026) and Berthold, Cesa-Bianchi, Di Pace, & Haberis (2024), among many others. The former two exploit the series to sharpen identification in their VAR models, while the latter three use it in other applications. Känzig (2023) finds that a carbon policy shock reduces emissions and economic activity. Bauer et al. (2026) bring the high-frequency approach to daily financial data, estimating event-study regressions of energy futures prices, market-based inflation expectations and nominal interest rates around EU ETS regulatory events. They find significant effects on energy markets and on inflation expectations, but no appreciable response of nominal rates. Their outcome variables and their frequency are the closest to ours in this literature, while their identification is the same as in Känzig (2023), still rests on the regulatory event window. To our knowledge, ours is the only other paper that identifies a shock to the supply of EUAs at daily frequency without relying on Känzig (2023). Rather than isolating shocks on a few event days, we recover an EUA supply shock series from a structural model, with one observation for each trading day. Since we compare the two shocks throughout the paper, we label the carbon policy shock of Känzig (2023) the *HF shock*, while we label our EUA supply shock the *SVAR shock*.

A small number of papers identify the aggregate effects of carbon pricing at lower frequencies. Metcalf & Stock (2023) estimate local projections on a panel of countries in the EU ETS and identify the effects of a carbon tax from the variation in carbon tax systems across those countries. They find no evidence that the tax adversely affects employment or GDP growth, and a modest reduction in emissions. These carbon taxes apply to emissions not covered by the EU ETS, and hence to sectors with high marginal abatement costs. Our approach is similar to Bjørnland et al. (2024), who study the European carbon market in a three-variable SVAR with emissions, the EUA price, and industrial production at monthly frequency. We

instead work at daily frequency and focus on financial variables. This allows us to identify EUA supply shocks from daily asset prices and to study their effects on both high- and low-frequency variables. In contrast to Metcalf & Stock (2023), Bjørnland et al. (2024) do find a substantial negative impact on economic activity and GHG emissions. Closely related, A. G. Gazzani & Taboga (2024) use a VAR model with data at daily and monthly frequency, and exploit shock-based restrictions to decompose EUA-price fluctuations into business-cycle, emission-intensity, allowance-supply, and market-sentiment shocks. Like us, they use narrative restrictions to identify the EUA supply shock. They ultimately rely, however, on external-variable constraints to identify the shock, requiring their monthly EUA supply shock to be positively correlated with the monthly series of Känzig (2023). Our more parsimonious identification instead uses the carbon price and utility stock valuations to recover an EUA supply shock. We then aggregate this dense daily measure and use it as an instrument to study its effects on emissions and macroeconomic activity.

Känzig & Konradt (2023) disentangle the effects of carbon taxes and the carbon market by combining the approaches of Känzig (2023) and Metcalf & Stock (2023). They find that both policies reduce emissions, but that the carbon market carries higher economic costs, and they point to sectoral coverage as a key factor behind the difference. The power sector plays a central role here. Carbon taxes generally exclude it, while the EU ETS does not. This contrast partly motivates our identification restriction.

From a methodological perspective, several studies estimate an SVAR on high-frequency data and use an aggregate of the high-frequency shocks as instruments in lower-frequency proxy SVARs. Alessandri, Gazzani, & Vicondoa (2023) lay the foundation for this approach and identify the effects of uncertainty shocks. Brandt, Saint Guilhem, Schröder, & Van Robays (2021) study the effects of monetary policy, macro news and global risk shocks, A. Gazzani, Venditti, & Veronese (2024) identify global shocks using oil prices, while Adolfsen et al. (2026) study gas market shocks.

3 From EU ETS Installations to Utility Stocks

The EU ETS regulates emissions at the installation level, while our identification relies on the stock prices of listed utility companies. This link is direct because utilities own fossil-fuel power plants. These institutional features plausibly make utilities the sector most exposed to EUA supply shocks and motivate our focus on utility stocks.

Launched in 2005, the EU ETS is among the largest carbon markets in the world. It covers electricity and heat generation, energy-intensive industry, and aviation, which together account for around 40% of EU GHG emissions. It operates in all EU member states as well as Iceland, Liechtenstein, Norway, and, for electricity generation, Northern Ireland. The relevant period for our analysis begins with Phase 3 (2013–2020) and continues thereafter. Phase 3 produced a stronger scarcity and price signal than the earlier phases, which were essential for establishing the system but relied much more heavily on the free allocation of EUAs. It introduced a single EU-wide cap in place of national caps, broadened the scope of coverage, and made auctioning the default method for allocating allowances.[1] Bauer et al. (2026) start their baseline sample in 2013 on the same grounds.

The EU ETS cap is expressed in EUAs. One allowance gives the right to emit one tonne of CO_2 equivalent. Covered firms must monitor and report emissions annually and surrender enough allowances to cover their verified emissions. Allowances can be allocated for free, purchased at auction, or traded in the market. For electricity generation, auctioning has been the main allocation method since 2013. This makes EUA prices directly relevant for utilities that own or operate fossil-fuel power plants.

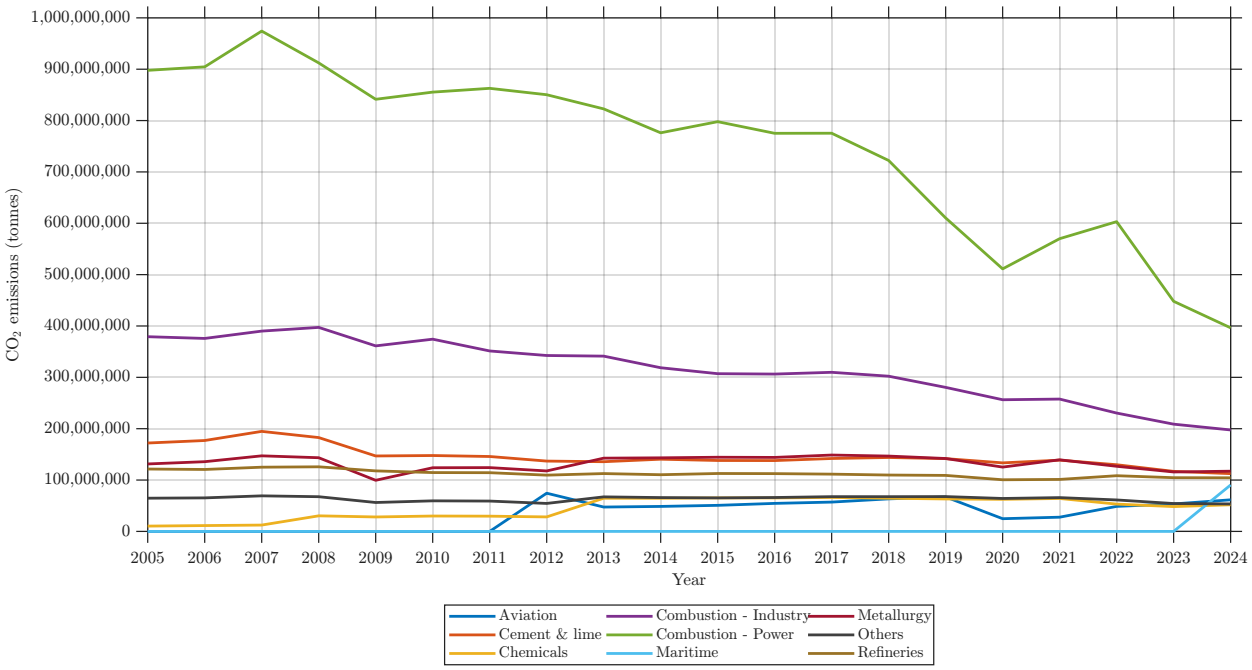
The EU ETS covers *activities* rather than financial sectors. An activity is a regulatory category, not a stock-market classification. An *installation* is a physical unit in which a firm carries out one or more regulated activities, and the operator of the installation is responsible for compliance. Installations are therefore the regulated entities, while listed stocks are financial claims on parent companies that may own or control many regulated and unregulated subsidiaries. The European Union Transaction Log (EUTL) reports transactions and compliance information, with a delay, for installations covered by the EU ETS (European Commission, 2026b).

To show how emissions from different activities evolved over time, we use data from the Bruegel ETS Tracker (Bruegel, 2026), which brings together and harmonizes data from the EUTL and complementary sources. To make the many ETS activity types more manageable, the Tracker groups them into nine categories. Figure 1 shows how verified emissions evolved across these categories. The figure shows that Combustion - Power accounts for the largest share of covered emissions. The Bruegel ETS Tracker further reveals that utility firms operate the vast majority of installations in this category.[2]

[1] See European Commission (2026a) for further details.

[2] We do not link the index constituents to their installations. The EUTL identifies installations and account holders but rarely records the owning company, so the mapping requires matching against external company registers (Abrell, 2024; EUETS.INFO, 2026). The mapping also stays incomplete. Lynch & O'Hagan-Luff (2024) match installations to listed firms and reach 123 companies, a quarter of all installations and 45% of verified emissions in 2020.

Figure 1: Verified EU ETS Emissions by Activity Category



Notes: The lines show annual verified emissions covered under the EU ETS by activity category, measured in tonnes of CO_2. The sample covers 2005–2024; aviation enters in 2012 and maritime transport in 2024 as their coverage under the system begins. The data are retrieved from Bruegel (2026).

To translate this regulatory exposure into a financial-market measure, we turn to stock indices for European utilities, whose constituent firms operate the vast majority of installations in the Combustion - Power category. STOXX classifies companies according to their primary source of revenue. Under the Industry Classification Benchmark (ICB) taxonomy, each stock is assigned to a single subsector, which maps into broader sectors, supersectors, and industries. STOXX provides two benchmark indices for the utility market: the EURO STOXX Utilities index and the STOXX Europe 600 Utilities index. Both are supersector indices and contain firms classified under the Utilities supersector, which comprises Electricity, Gas, Water and Multi-utilities, and Waste and Disposal Services (STOXX Ltd., 2026). We rely on the utility indices of STOXX for our analysis. Whenever our sample does not include post-Brexit observations, we will rely on the STOXX Europe 600 Utilities index, which does include British firms.

For utility stocks to be informative about EU carbon policy, the listed firms should also be economically exposed to the EU. Revenue data from 2013 to 2025 show that this condition is broadly satisfied. For most large European utilities, the share of revenues generated outside Europe is small, and revenues are concentrated in the firm's home country. There are exceptions. Iberdrola and National Grid have substantial non-European revenue exposure,

and Enel has operations in Brazil. Still, for the vast majority of utility stocks, revenues are concentrated in Europe. This limits the extent to which non-European regulation drives the utility index.

While demand for the services of European utility companies primarily comes from European consumers, utilities are more insulated from global commodity supply shocks than energy companies. According to the ICB classification, the Energy supersector includes subsectors such as Integrated Oil and Gas, Oil: Crude Producers, Oil Refining and Marketing, Pipelines, and Coal. Energy stocks therefore represent relatively upstream activities whose revenues depend strongly on global commodity prices and production conditions. Utilities operate further downstream, generating, distributing, and selling electricity or gas directly to households and firms. Regulatory grandfathering through "carbon leakage" is also less prominent in the utility sector, since utilities, unlike steel firms, do not rely on exports.

4 SVAR Identification of the EUA Supply Shock

We estimate a daily ten-variable SVAR for European energy and financial markets. We rely on sign and narrative restrictions for identification. We identify the shock as one that raises the carbon price and lowers utility stocks on impact. To sharpen identification we also impose the sign of the shock on a small set of dates.

Let $\boldsymbol{y}_t$ be an $N \times 1$ vector of endogenous variables and let $\boldsymbol{x}_t$ be an $M \times 1$ vector of exogenous variables. We assume that the endogenous variables evolve at daily frequency according to a VAR(p) model:

$$\boldsymbol{y}_t = \boldsymbol{a} + \boldsymbol{b}t + \boldsymbol{A}_1\boldsymbol{y}_{t-1} + \cdots + \boldsymbol{A}_p\boldsymbol{y}_{t-p} + \boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{u}_t, \tag{1}$$

where $\boldsymbol{a}$ is a vector of constants, $\boldsymbol{b}$ a vector of trend coefficients, $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_p$ and $\boldsymbol{C}$ are coefficient matrices and $\boldsymbol{u}_t$ is the vector of reduced-form innovations.

We assume the reduced-form innovations to be linear combinations of structural shocks:

$$\boldsymbol{u}_t = \boldsymbol{B}\boldsymbol{e}_t, \tag{2}$$

where $\boldsymbol{e}_t$ denotes the vector of structural innovations and $\boldsymbol{B}$ is the contemporaneous impact matrix.

The model includes ten endogenous variables, which capture the dynamics of European equity and energy markets. We impose the sign restrictions on the first two: the carbon

price and the STOXX Europe 600 Utilities index. Five further variables account for broader European financial-market dynamics: the STOXX Europe 600 index, the VSTOXX volatility index, the two-year and the ten-year German government bond yields, and the STOXX Europe 600 Energy index. The remaining three are coal, gas, and oil, the key commodities for the European energy market. All commodities, including the carbon price, enter as front-month futures prices. The vector $\boldsymbol{x}_t$ contains one exogenous variable, the MSCI World index, which enters contemporaneously and with three lags. The identifying assumption is that news about the supply of European allowances does not move global equity within the day, while global disturbances do. Under that assumption the world index is predetermined with respect to the EUA supply shock, and conditioning on it contemporaneously removes the variation explained by global equity markets from the reduced-form innovations. The sign restrictions then select among innovations specific to European markets rather than among innovations that still carry a common global component. We measure the index by its German-listed iShares ETF, so that the control trades in the same window as the endogenous series.

All assets therefore trade on European exchanges and their trading hours broadly overlap, though the overlap is not exact, since trading hours vary slightly across Europe. We convert the series quoted in other currencies into euros. All variables enter in log-levels, except the two bond yields, which enter in levels. We include a constant and a linear time trend as deterministic variables and, like Alessandri et al. (2023), set the lag length to three for the endogenous variables. We estimate the model with Bayesian methods, using a flat Jeffreys prior.[3] To set-identify the model, we draw until we obtain 10000 draws that satisfy the imposed restrictions. The shock series starts with Phase 3 of the EU ETS (Section 3) on 1 January 2013. Appendix A collects details on the data.

We include a large number of endogenous variables to ensure invertibility. Invertibility requires that the estimated residuals span the structural shock of interest (Alessandri et al., 2023). Economically, allowance prices respond within the day to fuel, power, equity, and fixed-income news, and the compliance costs of the covered utilities depend jointly on gas and coal prices and on the two points of the term structure that discount their abatement investment. A smaller VAR may not be informationally sufficient. We test informational sufficiency in Section 5, once the shock is aggregated to monthly frequency. The number of endogenous variables is comparable to the daily models of Alessandri et al. (2023), which include between six and sixteen series.

[3]For implementation we rely on the toolbox of Ferroni & Canova (2021) and refer to it for the details.

We identify the shock with sign and narrative restrictions. A positive EUA supply shock has a positive effect on the carbon price and a negative effect on the utility index on impact.

Almost every activity covered by the EU ETS emits CO_2, so fluctuations in allowance demand are as broad as the covered economy itself. Few disturbances, by contrast, raise the carbon price and lower utility valuations at the same time. News that lowers the expected supply of allowances does. Such news raises the carbon price, and it raises both the compliance cost and the transition risk of firms that operate fossil-fuel generation, which lowers their valuation. Disturbances that work through allowance demand do not. When demand for the output of European utilities rises, utility valuations rise, while the higher demand for fossil-fuel inputs raises allowance demand and with it the carbon price. Demand-type disturbances move the two series in the same direction, and the restriction discards them.

Our identification rests on the assumption that a rise in the price of an input that is essential to utilities lowers their valuation. Carbon allowances are such an input, but so are oil and gas. A rise in the global price of oil or gas raises the input costs of European utilities and lowers their profitability, while it also raises the carbon price through the relative cost of gas- and coal-fired generation. The sign restrictions alone do not rule out such energy price disturbances. Bauer et al. (2026) document spillovers in the opposite direction, from carbon prices to energy futures, so energy prices cannot be treated as predetermined with respect to the shock.

Therefore, to sharpen identification, we impose the sign of the EUA supply shock on a set of dates (Antolín-Díaz & Rubio-Ramírez, 2018). We are conservative in choosing them. First, we rely on only four independent events for the sample ending in 2019. By choosing independent events we address the fact that the same policy measure is often the center of attention on multiple days. Second, we rely on dates explicitly discussed by Känzig (2023) or dates used by A. G. Gazzani & Taboga (2024) for narrative identification of their EUA supply shock.

Table 1 collects the events, together with the imposed sign and the academic literature that lends support to these restrictions. To illustrate we briefly discuss two events. First, on 16 April 2013 the European Parliament surprised markets by voting against a proposal that would have reduced the supply of allowances. The Economist ran the headline "ETS, RIP?". This is the type of regulatory news used to construct the carbon policy surprise series and explicitly discussed in Känzig (2023). Second, on 15 February 2019 the European Commission adopted a more selective carbon-leakage list for the fourth phase of the EU ETS (2021–2030). The list determines the allocation of free allowances to industry and was unexpectedly restrictive. Like A. G. Gazzani & Taboga (2024) we impose that the sign of

the EUA supply shock is positive on that day. Känzig (2023) excludes this event because of potential contamination from other shocks, namely "bearish economic data and worries ahead of Brexit deadline". Our model accounts for multiple shocks while identifying only one, so we can include events that a high-frequency design has to discard.

Table 1: Narrative Restrictions

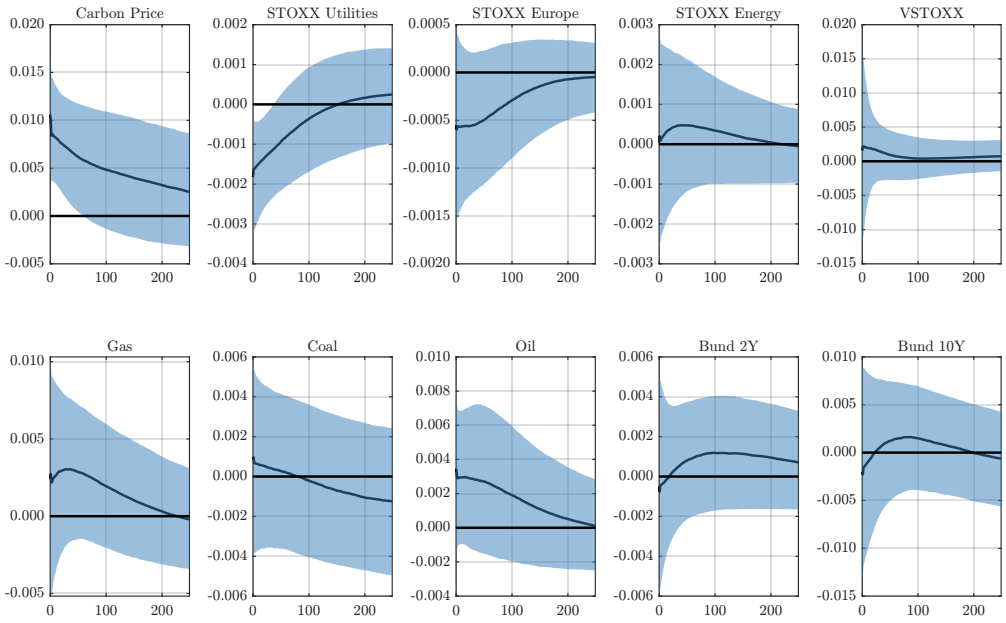
Date	Event	Sign	Source
01 Feb 2013	Carbon Market Forum background document points to EUA oversupply	+	GT
16 Apr 2013	EP votes *against* back-loading	−	K (explicit), GT
15 Feb 2019	Carbon leakage list for 2021–2030 narrowed	+	GT, K$^{\times}$
28 Aug 2019	Commission amends phase-IV auctioning regulation	+	K (explicit)

Notes: The table lists the four dates on which we impose the sign of the EUA supply shock, over the 2013–2019 daily sample. A + denotes a tightening of allowance supply, which raises the carbon price and lowers the utility index; a − denotes a loosening. K = Känzig (2023); GT = A. G. Gazzani & Taboga (2024); K$^{\times}$ = in his Table A.1 but *discarded* (#116).

Figure 2 presents the responses to an EUA supply shock. The black lines display the median response, while the blue shades show the 68% credible sets. A one-standard-deviation EUA supply shock raises the carbon price and lowers the utilities index. The responses of the remaining variables are estimated with substantial uncertainty: the credible set always includes zero, and many of the median responses are muted. The median responses nevertheless suggest a fall in the broad stock market index, which is smaller than the fall in the utilities index, and a rise in oil and gas prices. A negative response of the stock market and a positive response of oil prices are consistent with the monthly impulse responses to a carbon policy shock in Känzig (2023). Gas prices rise, while the response of coal prices is muted and changes sign over the horizon, in line with a coal-to-gas switch. As emission costs rise, electricity generation from gas becomes cheaper relative to coal. The pattern is consistent with Bauer et al. (2026), who find significant spillover effects from carbon prices to the natural gas market and oil markets at daily frequency. The positive effect of a carbon policy shock on oil prices is a consistent finding in the literature. It can be attributed to the carbon market also covering oil producers and refineries (see Känzig & Konradt (2023) or Bauer et al. (2026)). The muted response of the bond yields is again consistent with Bauer et al. (2026), who find no significant response of nominal interest rates to carbon policy surprises. Like our median responses, their point estimates are generally positive but fall well short of statistical significance.

Our goal is to construct a single EUA supply shock series. The sign and narrative restrictions set-identify the model. Each accepted draw delivers its own EUA supply shock series. Selecting one model would discard the information in all remaining draws. A natural summary

Figure 2: Daily Responses to an EUA Supply Shock



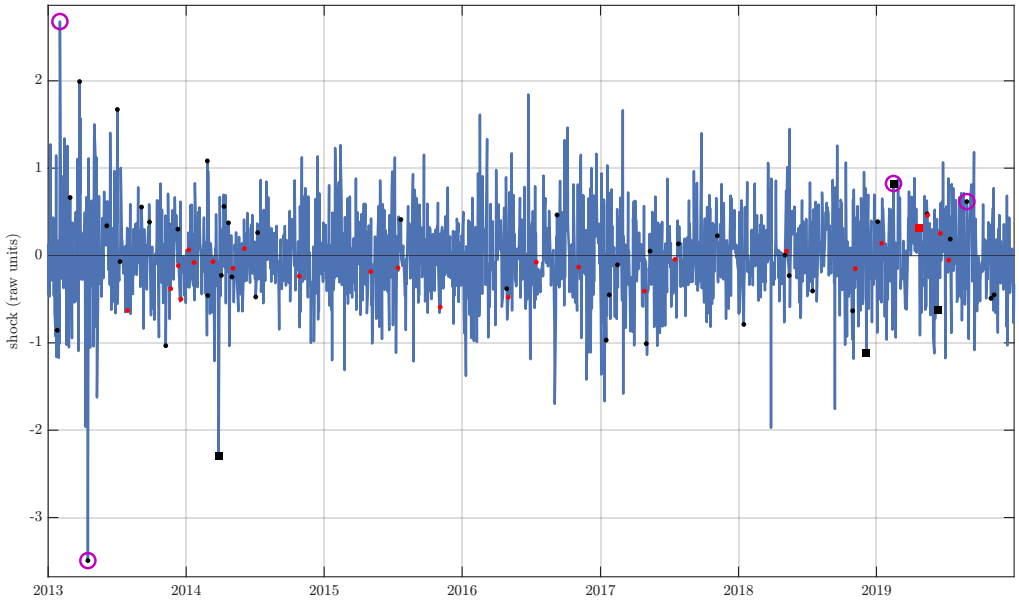
Notes: Responses to an EUA supply shock, identified with sign and narrative restrictions. The black lines display the pointwise posterior median responses. The blue bands report the 68% credible sets.

is the pointwise posterior median, that is, the median of the shock across accepted draws at each date. It is the shock of no admissible model, and its standard deviation is not one. We call it the SVAR shock.

Figure 3 presents the EUA supply shock at daily frequency and compares it to the carbon policy shock of Känzig (2023). We label the two measures SVAR and HF, respectively. Our SVAR shock has one observation per trading day, whereas the HF shock has 64 non-zero observations in the 2013–2019 overlap used for this comparison. To facilitate comparison, we therefore focus on those 64 announcement days and compare the HF series to the daily SVAR shock on the same dates. These days are highlighted by dots. We additionally include five days that Känzig (2023) excludes because they might be confounded by other macroeconomic news. Such days are highlighted by squares. While there are further excluded days we are only confident in attaching an interpretation to those five. Black and red represent agreement and disagreement in sign, with the HF shock on announcement days and with our reading of the regulatory action on the excluded days. Purple circles indicate the days on which we impose the sign of the EUA supply shock. For example, the largest negative value in the sample carries a black dot and a circle. The black dot indicates sign agreement with the HF series, and the purple circle shows that we imposed the sign.

On the 64 announcement days, the daily SVAR shock and the HF surprise correlate at 0.70, and their signs match on 63% of the days. Since the carbon policy surprise is constructed through careful narrative analysis, the positive correlation lends support to our shock series capturing EUA supply shocks.

Figure 3: Daily EUA supply shocks



Notes: The blue line presents the daily SVAR EUA supply shock over the 2013–2019 sample. Dots represent the 64 announcement days of Känzig (2023). Squares represent the five days excluded by Känzig (2023) due to potentially confounded events. Black indicates sign agreement with the HF shock, red indicates sign disagreement. Purple circles represent the four days on which the narrative restrictions impose the sign of the shock.

In sum, the daily model delivers one EUA supply shock realization for every trading day, and two pieces of evidence support its interpretation. The estimated responses are consistent with the pattern Bauer et al. (2026) document with high-frequency event studies. Also, the shock series is positively correlated with the surprise series of Känzig (2023). Emissions and economic activity are not observed at daily frequency, so the daily responses cannot speak to the aggregate effects of the shock. We therefore aggregate the series to monthly frequency and use it as an external instrument in a monthly proxy-SVAR.

5 The Aggregate Effects of EUA Supply Shocks

We study the aggregate effects of EUA supply shocks on emissions and the European macroeconomy. Consistent with the literature, we find that an EUA supply shock lowers emissions and economic activity. To show this, we replicate the proxy-SVAR in Känzig (2023) and compare the responses obtained with our SVAR EUA-supply instrument to those from the HF carbon policy surprise. The model is a standard proxy-SVAR, estimated by OLS and identified with a single external instrument (Mertens & Ravn, 2013; Stock & Watson, 2018). Appendix B gives a brief overview of the empirical specification.

Following Alessandri et al. (2023), we aggregate the daily shock values by averaging all trading-day observations within a given calendar month.[4] We compute the structural IRFs from three SVARs, which differ only in the instrument we use. First, we estimate the model using the HF instrument of Känzig (2023). Second, we estimate the same model using our SVAR instrument. The HF instrument runs from 2005 to 2019, while the SVAR instrument runs from 2013 to 2019. Third, we estimate the model using a censored version of our instrument, which we label SVAR^0. At daily frequency, SVAR^0 equals the SVAR shock on every day on which the HF shock is zero, and is set to zero on all remaining days. The comparison between SVAR and SVAR^0 therefore isolates the role of announcement days in our daily shock. The censoring sets many of the largest values to zero, because they coincide with announcement days, as Figure 3 shows. If those days drive the results, the two IRFs should differ. If the IRFs overlap, the macro responses are coming from daily variation outside those announcement windows.

Figure 4 displays the impulse responses, normalized to increase HICP energy by 1 percent on impact. The solid blue lines show the responses for the SVAR instrument, together with the 68 and 90 percent confidence bands in blue. The dashed blue lines show the point estimates for the SVAR^0 instrument, while the red lines show the point estimates for the HF instrument.

A positive EUA supply shock, which tightens allowance supply, raises energy prices on impact, with the increase lasting for about two years, and lowers GHG emissions persistently. We measure the price elasticity of emissions as the maximum percentage decline in emissions following a 1 percent increase in energy prices. The estimated elasticity is around 0.59, close to the value of around 0.6 in Ortubai et al. (2025) and below the 0.75 of Känzig (2023).

[4]The monthly shock also lets us verify informational sufficiency. Following Alessandri et al. (2023), we regress it on the first lag of seven factors extracted from the US macro-financial panel of McCracken & Ng (2016) and apply the orthogonality test of Forni & Gambetti (2014). Over 2013–2019 the test does not reject, neither for our SVAR shock ($F = 1.23$, $p = 0.30$) nor for the HF surprise ($F = 1.43$, $p = 0.21$).

The responses of the macroeconomic variables show the economic cost of reducing emissions. Industrial production falls by around 0.7 percent, though the effect in the first months is estimated with considerable uncertainty. The unemployment rate rises by over 0.1 percentage points. Headline HICP increases by around 0.2 percent and, like HICP energy, stays elevated for almost two years. The two-year rate rises briefly. Stock prices drop by up to 2 percent, but this response is also estimated with considerable uncertainty. Oil prices increase on impact by over 5 percent. The EU ETS also covers oil producers and refineries, which makes such a response plausible.

The solid and dashed blue lines lie close together. Censoring the instrument on exactly the announcement days behind the HF measure therefore leaves the responses unchanged. Because allowances are bankable and priced on the expected path of scarcity, news about future supply need not arrive on a scheduled announcement day, and the daily variation outside those windows is informative about allowance supply.

The monthly responses resemble their counterparts at daily frequency displayed in Figure 2. In both cases the European stock market falls, oil prices rise, and interest rates display a hump-shaped response.

The red lines display the point estimates for the carbon policy shock of Känzig (2023). They lie inside the confidence bands of our EUA supply shock, with the exception of GHG emissions, which lie outside for the first few months. [5] Three differences arise. First, the SVAR shock has stronger effects on prices: HICP and HICP energy rise by more, and the rise in HICP is more persistent. Second, the effect on the real variables and on emissions is weaker: industrial production and GHG emissions fall by less, and the unemployment rate rises by less. Third, oil prices rise by more with the SVAR shock than with the HF shock. Energy price disturbances also satisfy our sign restrictions (Section 4), and residual contamination by such disturbances can contribute to the larger oil response. The responses of the two-year rate and of stock prices are nearly identical. Taken together, the SVAR shock moves prices by more and real variables by less than the HF shock, which is a lower price elasticity of emissions: 0.59 against 0.75.

The conditional correlation between the two monthly instruments, again conditioning on a non-zero HF-shock observation, is 0.26, while the signs of the two monthly shocks align in 53% of the months for which the monthly HF shock is non-zero. The structural shocks of the two proxy-SVARs correlate at 0.89, mirroring the similar impulse responses. While these values

[5] Appendix B presents the results for the reverse exercise, where the carbon policy surprise instrument is taken as the baseline model. The SVAR and SVAR^0 responses lie inside the HF bands for all variables and all horizons.

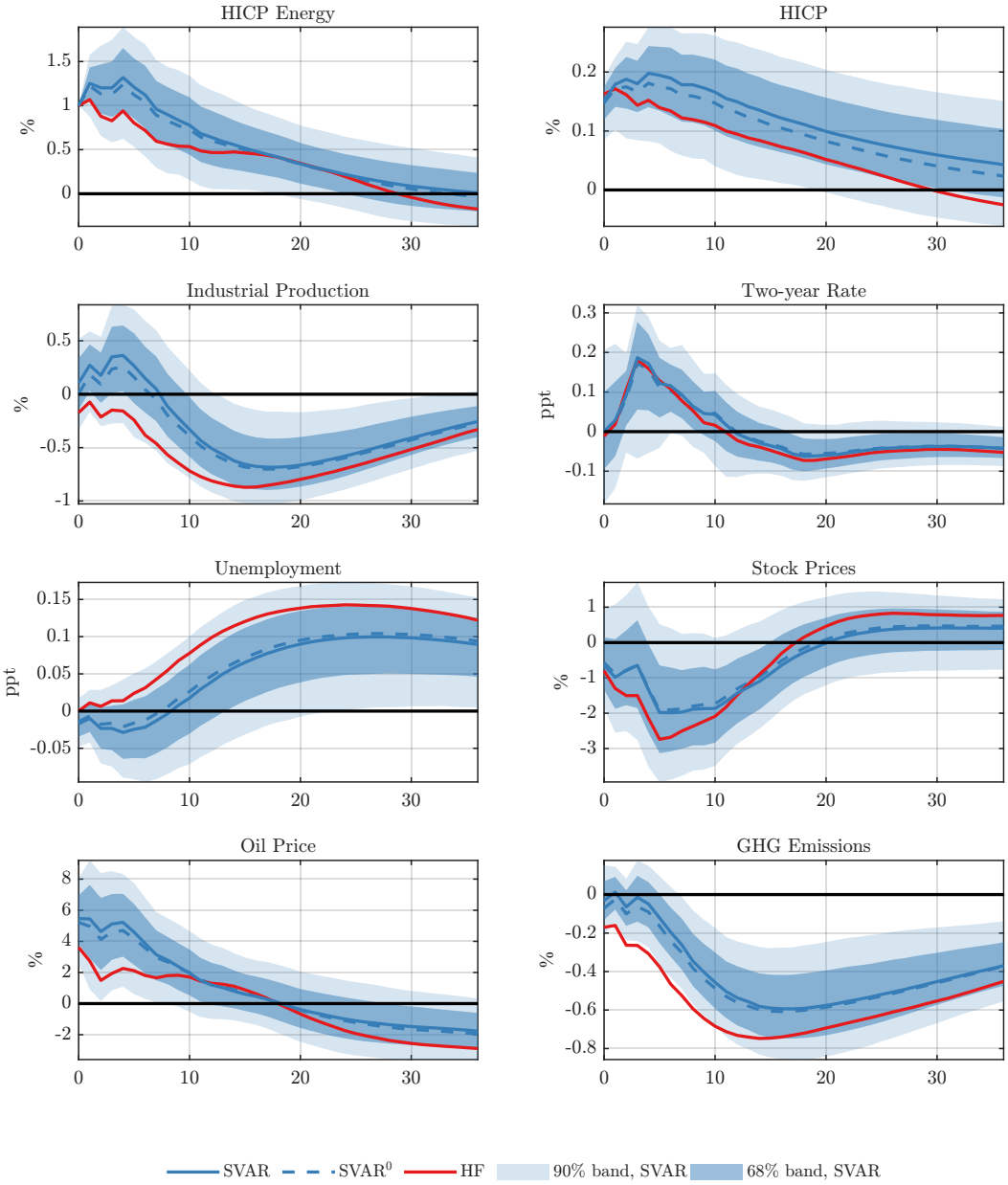
might seem low, Brennan, Jacobson, Matthes, & Walker (2025) find similar magnitudes when comparing various high-frequency monetary shock series, even though those shocks are constructed using the same identification strategy. Given that the HF carbon-policy shock and the SVAR EUA-supply shock are constructed using different methods, we view these magnitudes as consistent with the two series capturing overlapping carbon-market disturbances only imperfectly. Our restrictions admit any disturbance that tightens effective allowance supply, not regulatory news alone, and this broader coverage is a plausible source of differences reported above.

The robust F-statistic of the HF instrument, at 16.85, is larger than the value of 12.67 for the SVAR instrument and 13.02 for the SVAR^0 instrument. This is partly due to the HF instrument being longer, as it starts in 2005 rather than 2013. Truncating the HF instrument to run from 2013 to 2019 yields a robust F-statistic of 7.00. This suggests that part of the instrument strength of the HF instrument reflects sample length rather than identification alone.[6]

In sum, the two instruments deliver similar responses, with the SVAR shock implying a somewhat lower price elasticity of emissions. Even though the EUA supply shock captures a broader set of disturbances than the carbon policy shock, which isolates regulatory news about the supply of allowances, the two agree on the sign, the shape and the persistence of the effects on the macroeconomy and emissions. SVAR and SVAR^0 deliver almost identical responses. We conclude that EUA supply shocks do not occur only on policy-announcement days. They appear at a higher frequency, and the daily SVAR recovers this broader variation. At monthly frequency, this advantage is small, as the HF shock has a non-zero observation in the vast majority of months. The SVAR shock might, however, prove useful for many data-hungry applications at daily frequency, for which the HF instrument is too sparse.

[6]The weak-instrument robust bands of Montiel Olea, Stock, & Watson (2021) are not available here: their covariance estimator requires $n^2p + n(n+1)/2 + nK < T$, which at $n = 8$, $p = 6$ and $K = 1$ means 428 parameters against $T = 246$ observations.

Figure 4: Impulse Responses to a EUA Supply Shock

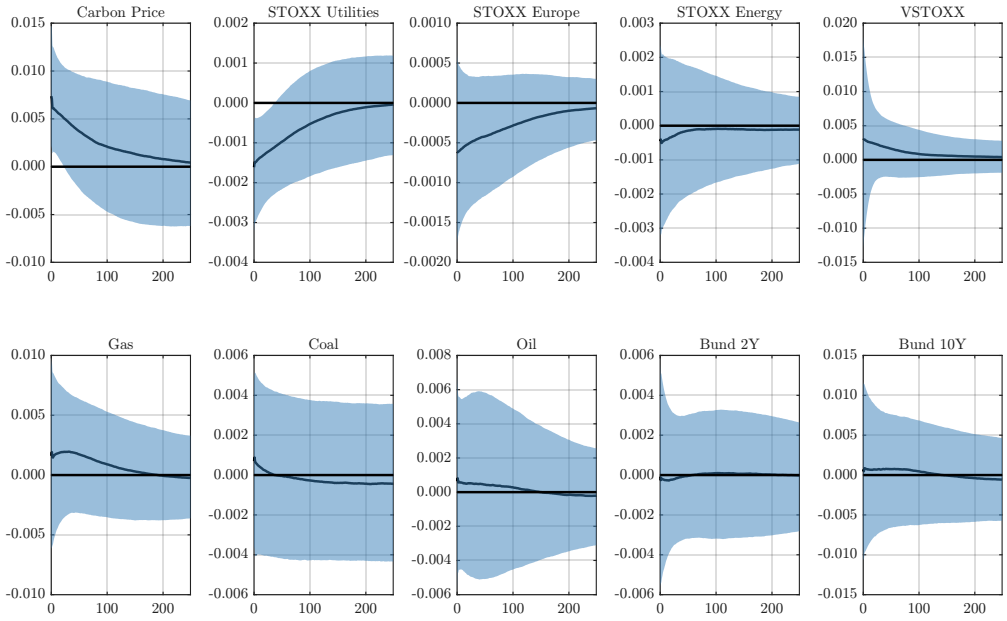


Notes: Impulse responses estimated with the SVAR EUA-supply instruments and the HF carbon-policy instrument, normalized to increase HICP energy by 1 percent on impact. Solid blue line: point estimate from the SVAR instrument. Dashed blue line: point estimate from the SVAR^0 instrument. Solid red line: point estimate from the HF instrument. Dark and light blue shaded areas: 68 and 90 percent confidence bands based on the moving block bootstrap. The VAR is estimated with six lags.

6 Robustness to the Narrative Restrictions

We ask how much of the identification the sign restrictions deliver on their own. We repeat the estimation of Section 4 without imposing the narrative restrictions. Figure 5 presents the impulse responses. The point estimates are similar for most variables, with the exception of the commodities, which now show a muted response. The credible sets are considerably wider. The agreement with the findings of Bauer et al. (2026) disappears.

Figure 5: Daily Responses to an EUA Supply Shock: Sign Restrictions Only

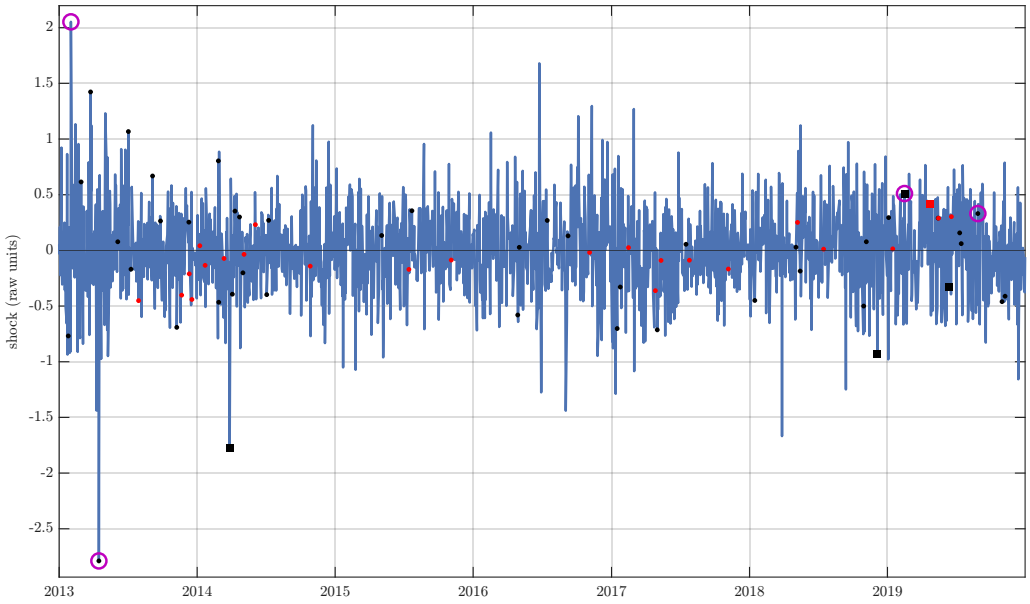


Notes: Responses to an EUA supply shock, identified with sign restrictions only. The black lines display the median responses. The blue bands report the 68% credible sets.

Figure 6 presents the shock series, as in Figure 3. Conditional on an announcement day, the correlation with the carbon policy surprise series is 0.74 and the signs match on 64% of days, slightly above the 0.70 and 63% of the main specification. The purple circles indicate the four dates on which the main specification imposes the sign of the shock. The sign restrictions alone deliver the same signs on those dates.

Figure 7 presents the impulse responses to an EUA supply shock, identified only with sign restrictions in the daily model. The robust F-statistic is 10.9 for the SVAR instrument and 11.2 for the SVAR^0 instrument, below the 12.7 and 13.0 of the main specification. The responses are qualitatively similar to those of the main specification and are estimated with greater uncertainty. The point estimates from the HF and the SVAR instrument agree more

Figure 6: Daily EUA Supply Shocks: Sign Restrictions Only

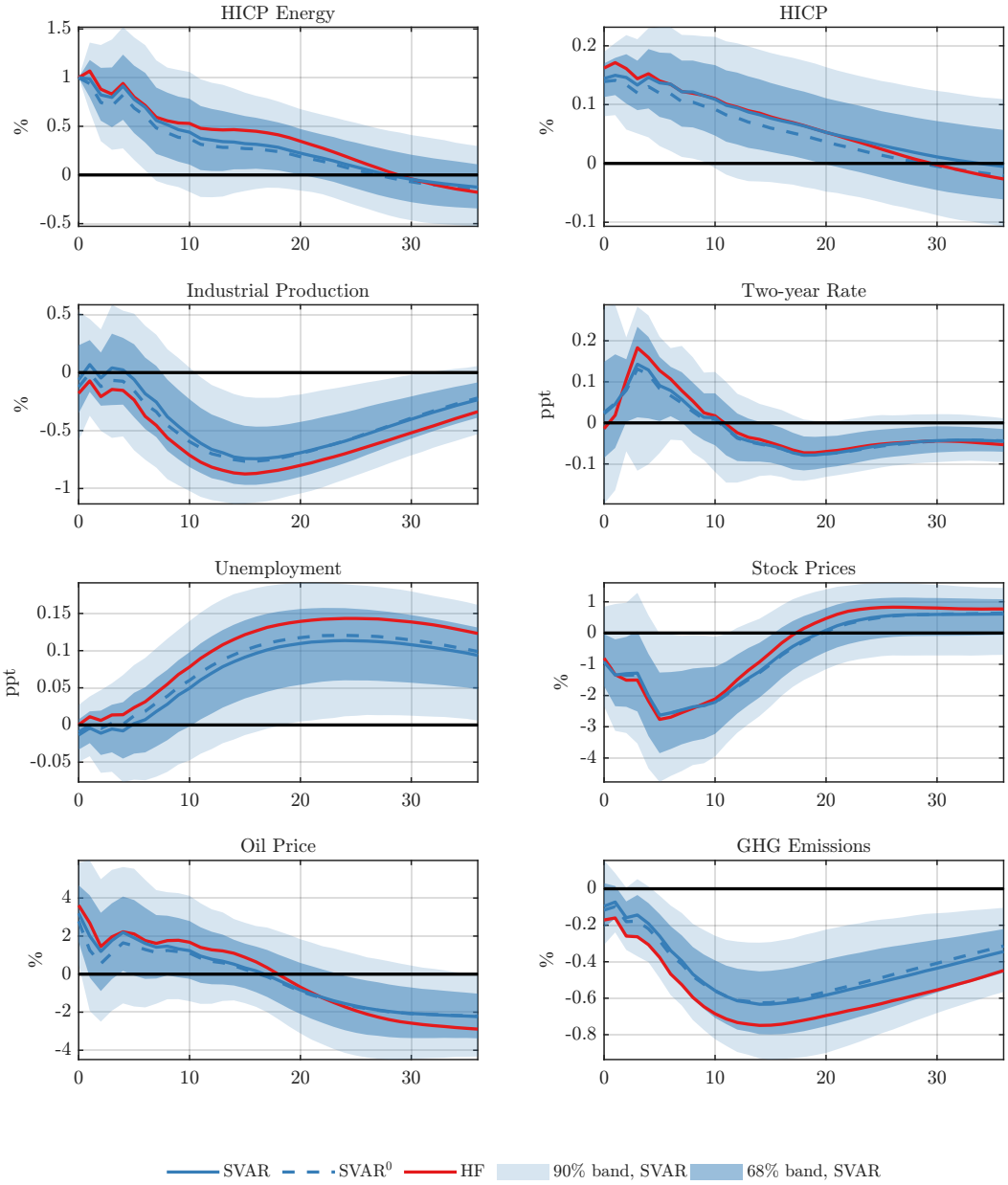


Notes: The blue line presents the daily SVAR EUA supply shock over the 2013–2019 sample. Dots represent the 64 announcement days of Känzig (2023). Squares represent the five days excluded by Känzig (2023) due to potentially confounded events. Black indicates sign agreement with the HF shock, red indicates sign disagreement. Purple circles represent the four days on which the narrative restrictions impose the sign of the shock.

closely than in the main specification and overlap in many instances. The correlation between the two proxy-SVAR shocks is 0.96.

In sum, the sign restrictions alone already go a long way in pinning down the EUA supply shock. Nevertheless, the narrative restrictions sharpen identification by narrowing the identified set of the daily model and deliver daily responses that align much closer to the findings of Bauer et al. (2026).

Figure 7: Impulse Responses Using HF and SVAR Instruments: Sign Restrictions Only



Notes: Impulse responses to an EUA supply shock identified from sign restrictions only. Solid blue line: point estimate from the SVAR instrument. Dashed blue line: point estimate from the SVAR^0 instrument. Solid red line: point estimate from the HF instrument. Shaded areas: 68 and 90 percent confidence bands.

7 Conclusion

We identify a shock to the supply of EU emission allowances (EUAs) on every trading day, using a ten-variable SVAR of European energy and financial markets with sign and narrative

restrictions. Aggregated to monthly frequency and used as an external instrument, the shock reproduces the trade-off between emissions and economic activity that the literature finds. An EUA supply shock that raises energy prices by 1 percent lowers greenhouse gas emissions by around 0.59 percent at peak and industrial production by around 0.7 percent, and raises the unemployment rate by over 0.1 percentage points.

The daily shock lines up with the high-frequency carbon policy surprise of Känzig (2023) where the two overlap. Conditional on an announcement day, the two series correlate at 0.70 and agree in sign on 63% of days, and the monthly proxy-SVAR delivers similar responses with either instrument, with our shock moving prices by more and real variables by less and implying a somewhat lower price elasticity of emissions. Censoring our shock on exactly the announcement days behind the high-frequency measure leaves the responses almost unchanged. Because allowances are bankable and priced on the expected path of scarcity, news about future supply need not arrive on a scheduled announcement day, and the daily variation outside those windows is informative about allowance supply.

The breadth of the measure is an important qualification. Our restrictions identify disturbances that tighten effective allowance supply relative to the abatement options available to utilities, not regulatory news alone, and energy price disturbances are the most plausible source of contamination. Four pieces of evidence point to supply-side disturbances dominating the series. The daily responses agree with the event-study evidence in Bauer et al. (2026). The series is positively correlated with a surprise series built entirely from regulatory news. The monthly responses are close to those obtained with that surprise as the instrument. And the responses hold up even when we drop the narrative restrictions. The label therefore describes what dominates the series rather than the full identified set. At monthly frequency the two measures are close substitutes. Therefore, the daily series can be useful where the high-frequency surprise is too sparse.

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A Data

Table 2 lists the daily series that enter the SVAR of Section 4, and Figure 8 plots them in levels. Table 3 lists the monthly series that enter the proxy-SVAR of Section 5.[7]

Table 2: Daily Data

Series	Code	Units	Source
EUA front-month futures price	LEXC.01	EUR/tCO_2	Datastream
STOXX Europe 600 Utilities	S2SEU2E	Index, EUR	Datastream
STOXX Europe 600	DJSTOXX	Index, EUR	Datastream
STOXX Europe 600 Energy	S2SEO2E	Index, EUR	Datastream
VSTOXX volatility index	VSTOXXI	Index	Datastream
Two-year German government bond yield	TRBD2YT	Percent p.a.	Datastream
Ten-year German government bond yield	TRBD10T	Percent p.a.	Datastream
TTF base-load natural gas front-month futures	ETMC.01	EUR/MWh	Datastream
Rotterdam coal front-month futures	LMCC.01~E	EUR/t	Datastream
Brent crude oil front-month futures	LLCC.01~E	EUR/bbl	Datastream
iShares Core MSCI World UCITS ETF	D:EUNL~E	EUR per share	Datastream
HF carbon policy surprise		Standardized	Känzig (2023)

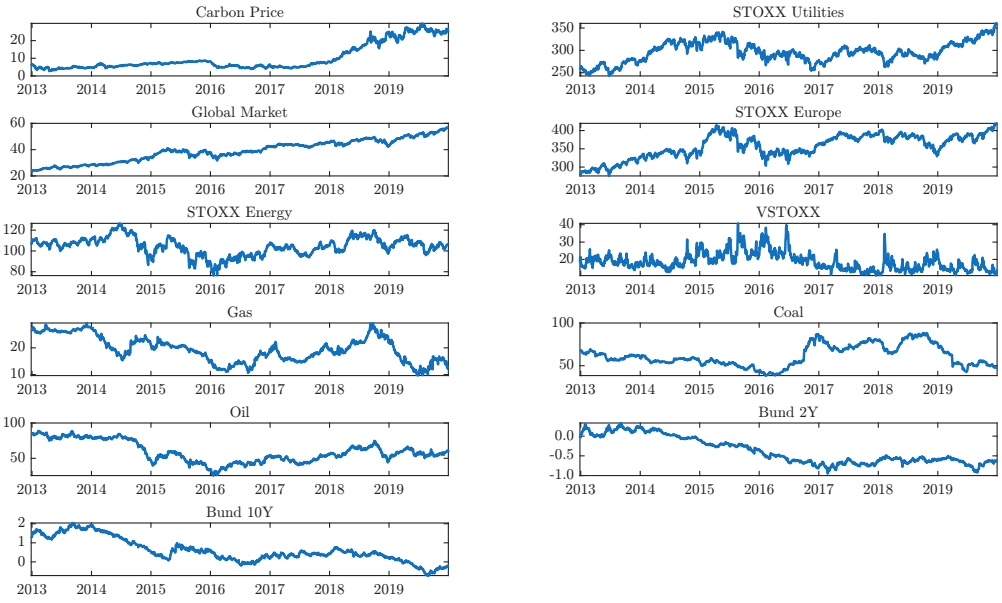
Notes: The daily sample runs from 3 December 2012 to 31 December 2019 and covers 1847 weekdays. The shock series starts on 1 January 2013 and covers 1826 weekdays. The first ten series are the endogenous variables of the SVAR, the MSCI World ETF is the exogenous control, and the high-frequency surprise of Känzig (2023) does not enter the model. Datastream converts the series quoted in other currencies into euros, marked by the suffix ~E.

Table 3: Monthly Data

Series	Code	Source	Sample
HICP energy (EA-19)	EKESCPENF	Känzig (2023)	Jan 1999 – Dec 2019
HICP all items (EA-19)	EKCPHARMF	Känzig (2023)	Jan 1999 – Dec 2019
GHG emissions	env_air_gge	Känzig (2023)	Jan 1999 – Dec 2019
Industrial production excluding construction (EA-19)	EKIPTOT.G	Datastream	Jan 1999 – Dec 2019
Two-year government bond yield	EMECB2Y.	Känzig (2023)	Jan 1999 – Dec 2019
Unemployment rate (EA-19)	EKESUNEMO	Känzig (2023)	Jan 1999 – Dec 2019
Euro STOXX	DJSTOXX	Känzig (2023)	Jan 1999 – Dec 2019
Brent crude price	DCOILBRENTEU	Känzig (2023)	Jan 1999 – Dec 2019

[7] We are deeply thankful to Diego Känzig for making the data available.

Figure 8: Daily Financial Data



Notes: The panels show the eleven raw daily series of Table 2 in levels, in the units given there, over the 2013–2019 sample.

B Replication of Känzig (2023)

We briefly describe the empirical specification of Känzig (2023). The VAR contains eight variables: the energy component of the Harmonised Index of Consumer Prices (HICP), headline HICP, industrial production, the two-year rate, the unemployment rate, a stock market index, the Brent crude oil price (the latter two deflated by the HICP), and total GHG emissions.[8] All variables enter in log-levels, except for the unemployment rate and the two-year rate, which enter in levels.[9] The sample runs from January 1999 to December 2019. Since our instrument runs from January 2013 to December 2019, we set the pre-2013 observations of the monthly shock series to zero. We estimate the VAR by OLS with six lags and include a dummy variable for the period from July 2011 to March 2012. Identification then proceeds in proxy-SVAR fashion. We project the reduced-form residual in the HICP energy equation on the external instrument and recover the relative impact vector from the remaining residuals. For inference, we use the residual-based moving block bootstrap and compute confidence bands from 10000 bootstrap replications. We also apply the small-sample

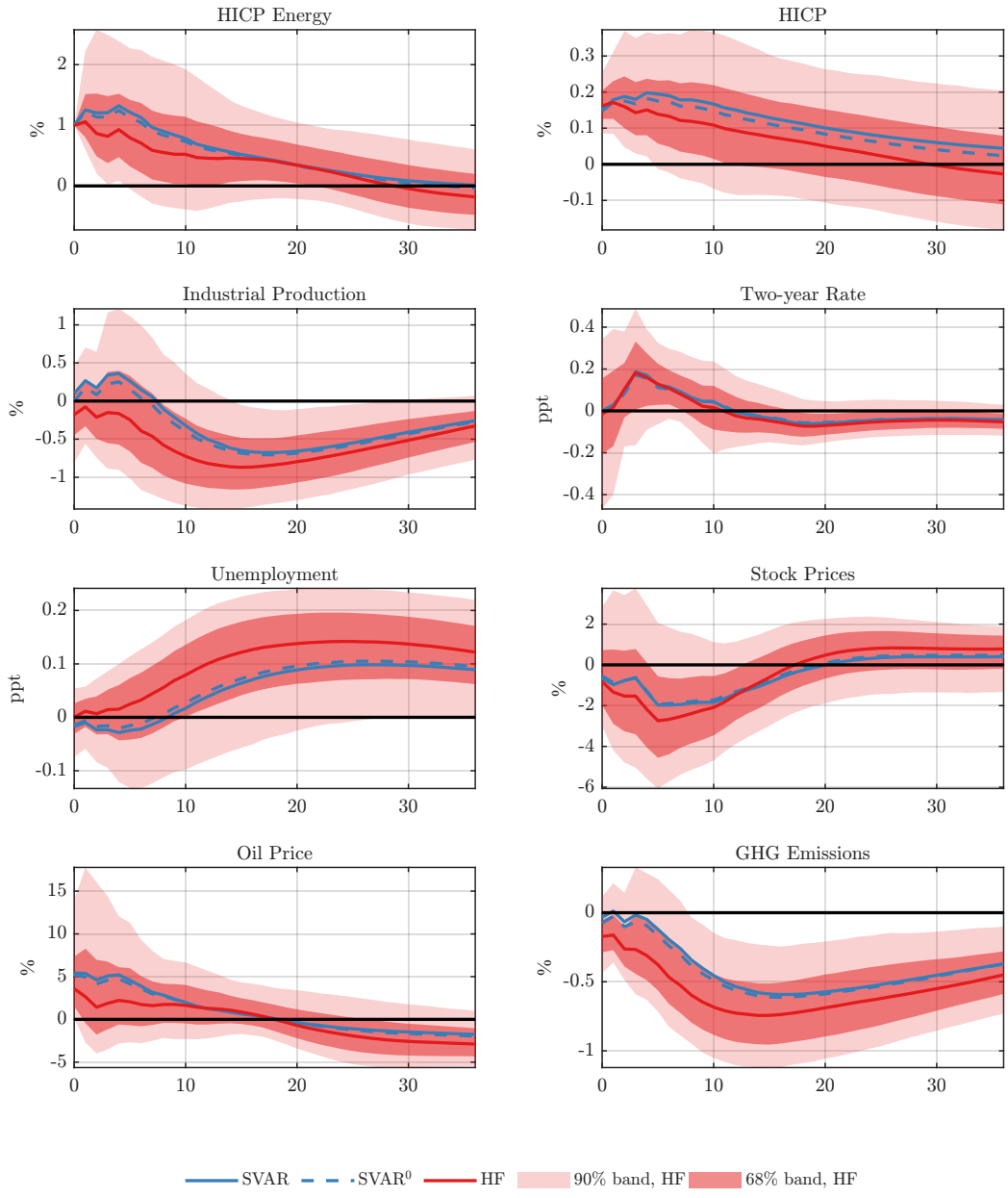
[8] We are deeply thankful to Diego Känzig for making the data available.

[9] The results are robust to a specification where the log-level variables enter in log differences.

bias correction of Hall (1992). We set the block length to 20, in line with the recommendation of Jentsch & Lunsford (2019) for similar sample sizes.

Figure 9 repeats the comparison of Figure 4 with the HF carbon policy surprise of Känzig (2023) as the baseline, so that the bands belong to that instrument rather than to ours. The red line and its confidence bands replicate Figure 3 of Känzig (2023). The point estimates from the SVAR and SVAR^0 instruments lie inside the 68 percent bands at every horizon and in every variable.

Figure 9: Impulse Responses Using HF and SVAR Instruments: HF Bands



Notes: Impulse responses estimated with the HF carbon-policy instrument and the SVAR EUA-supply instruments, normalized to increase HICP energy by 1 percent on impact. Solid red line: point estimate from the HF instrument. Dark and light red shaded areas: 68 and 90 percent confidence bands based on moving block bootstrap. Solid blue line: point estimate from the SVAR instrument. Dashed blue line: point estimate from the SVAR^0 instrument. Lag order: 6.